Regression Problem Set

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Problem 1 — Scatter Diagram and Linear Relationship

Context: A data analyst records the number of study hours (X) and the corresponding exam scores (Y) of 6 students. The data is shown below:

X (Hours Studied)	Y (Exam Score)
1	2
2	4
3	5
4	4
5	6
6	7

Tasks:

1. Draw the scatter diagram for this dataset.
2. Comment on whether a linear relationship seems appropriate between X and Y.

Problem 2 — Model Representation for Single Variable Linear Regression

Context: We are modeling the relationship between the number of hours studied (X) and the exam score (Y) using a single-variable linear regression model. The model assumes a straight-line relationship of the form:

$$\hat{y} = \beta_0 + \beta_1 x$$

where:

- β_0 is the intercept (value of Y when $X = 0$),
- β_1 is the slope (change in Y for each unit change in X).

The dataset is the same as in the previous problem:

X (Hours Studied)	Y (Exam Score)
1	2
2	4
3	5
4	4
5	6
6	7

Tasks:

1. Write the general model representation for single-variable linear regression.
2. Plot the dataset (scatter diagram) along with an example regression line $\hat{y} = 1 + x$.

Problem 3 — Single Variable Cost Function (MSE)

Context: We model the relationship between study hours (X) and exam score (Y) with a single-variable linear regression

$$\hat{y} = \beta_0 + \beta_1 x.$$

Use the same dataset:

X	Y
1	2
2	4
3	5
4	4
5	6
6	7

Let $m = 6$ denote the number of training examples.

Tasks:

1. Write the mean squared error (MSE) cost function $J(\beta_0, \beta_1)$.
2. Compute the cost for the following parameter choices: (i) $(\beta_0, \beta_1) = (0, 0)$, (ii) $(\beta_0, \beta_1) = (0, 1)$, (iii) $(\beta_0, \beta_1) = (2, 0.5)$.
3. Derive the partial derivatives $\dfrac{\partial J}{\partial \beta_0}$ and $\dfrac{\partial J}{\partial \beta_1}$.
4. (Visualization) Fix $\beta_0 = 0$ and plot $J(0, \beta_1)$ as a function of $\beta_1 \in [0, 2]$ to illustrate convexity of the cost.

Problem 4 — Least Squares Fit for Simple Linear Regression

Context: We want to find the best-fitting straight line for the dataset of study hours (X) and exam scores (Y), using the least squares method.

X	Y
1	2
2	4
3	5
4	4
5	6
6	7

Tasks:

1. Write the formulas for the least squares estimates of slope β_1 and intercept β_0:

$$\beta_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}, \qquad \beta_0 = \bar{y} - \beta_1\bar{x}.$$

2. Compute β_0 and β_1 for the given dataset.
3. Write the fitted regression line $\hat{y}$.
4. Plot the data points and the fitted line.

Problem 5 — Simple Linear Regression + Gradient Descent

Context: Predict exam **Score (Y)** from **Hours Studied (X)**.

Data (m = 5):

X (Hours)	Y (Score)
1	2
2	4
3	5
4	4
5	5

Tasks:

1. Fit simple linear regression model : $\hat{Y} = \beta_0 + \beta_1 X$.
2. Compute slope: $\beta_1 = \dfrac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$.
3. Compute intercept: $\beta_0 = \bar{y} - \beta_1\bar{x}$.
4. Predict the score for $X = 6$.
5. Perform one batch Gradient Descent step starting from $\beta_0 = 0, \beta_1 = 0$ with learning rate $\alpha = 0.1$:

   $$J(\beta_0, \beta_1) = \frac{1}{2m}\sum(\hat{y}_i - y_i)^2, \quad \beta_0 \leftarrow \beta_0 - \alpha \cdot \frac{1}{m}\sum(\hat{y}_i - y_i), \quad \beta_1 \leftarrow \beta_1 - \alpha \cdot \frac{1}{m}\sum(\hat{y}_i - y_i)x_i$$

   Compute updated β_0, β_1.

Problem 6 — Gradient Descent for Single Variable Linear Regression

Context: We want to minimize the cost function for linear regression using **batch gradient descent**. The dataset (number of hours of exercise X vs fitness score Y) is:

X	Y
1	1.5
2	1.9
3	3.2
4	3.0
5	4.1

Tasks:

1. Write the hypothesis function and cost function.
2. Derive the gradient descent update rules for β_0 and β_1.
3. Perform one iteration of gradient descent starting from $\beta_0 = 0, \beta_1 = 0$ with learning rate $\alpha = 0.1$.
4. Plot the dataset along with the regression line after this update.

Problem 7 - Linear Regression via Normal Equation

Context: We want to predict the final exam **Score (Y)** of a student from the number of **Study Hours (X)**.

Data (m = 3):

X (Hours)	Y (Score)
1	2
2	4
3	6

Tasks:

1. Fit the model
$$\hat{Y} = \beta_0 + \beta_1 X.$$
2. Write the model in matrix form: $\hat{Y} = X\beta$.
3. Use the Normal Equation:
$$\hat{\beta} = (X^\top X)^{-1} X^\top Y$$
to compute β_0, β_1.
4. Predict the score when a student studies for 4 hours.

Problem 8 - Linearity Assumption

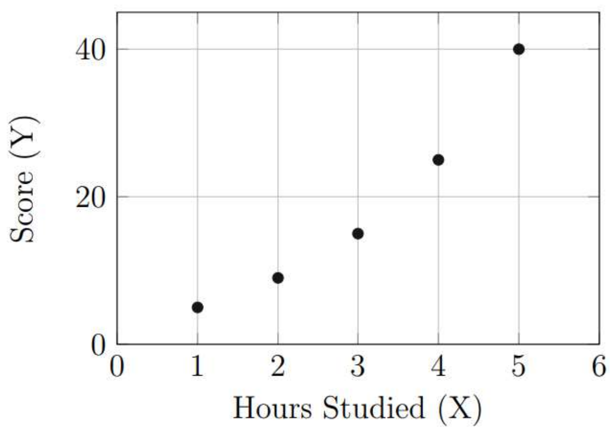
Context: Suppose we are modeling the relationship between **Study Hours (X)** and **Exam Score (Y)** using linear regression. A small dataset is collected:

X (Hours)	Y (Score)
1	5
2	9
3	15
4	25
5	40

Tasks:

1. Plot the data points.
2. Discuss whether the assumption of **linearity** (i.e., Y increases in a straight-line manner with X) seems valid.
3. If linearity does not hold, suggest a possible transformation of X that might make the relationship more linear.

Plot of the Data



Problem 9 - Homoscedasticity Assumption

Context: We fit a regression model predicting **Salary (Y, \$1000)** from **Years of Experience (X)**. The residuals (prediction errors) from the fitted model are:

$$\text{Residuals} = \{-2,\ -1,\ 0,\ 1,\ 5\}.$$

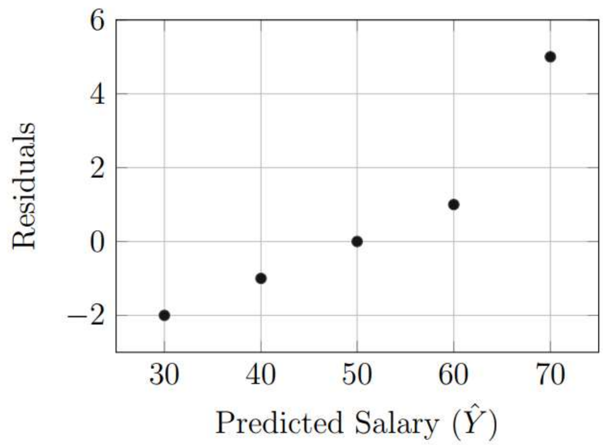
The corresponding predicted salaries were:

$$\hat{Y} = \{30,\ 40,\ 50,\ 60,\ 70\}.$$

Tasks:

1. Plot residuals versus predicted values.
2. Does the assumption of **constant variance (homoscedasticity)** seem to hold?
3. Explain why violation of this assumption can cause problems for regression.

Residual Plot



Problem 10 - Properties of the Regression Line

Context: We want to study the relationship between **Study Hours (X)** and **Exam Score (Y)** using linear regression.

Data (m = 4):

X (Hours)	Y (Score)
1	2
2	3
3	5
4	4

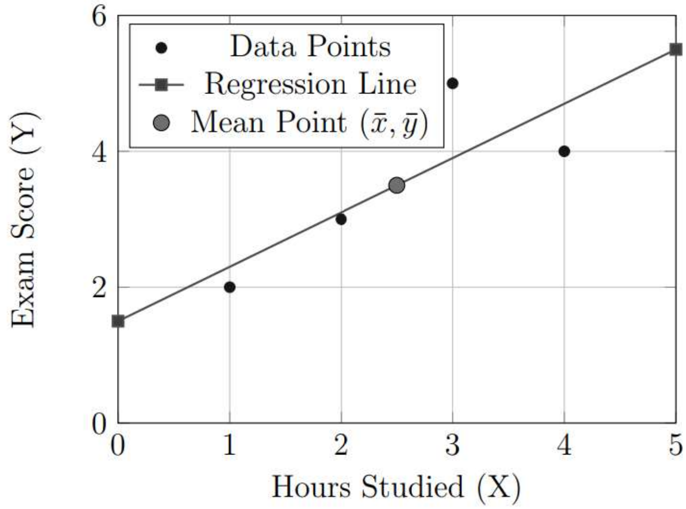
Tasks:

1. Fit the regression line: $\hat{Y} = \beta_0 + \beta_1 X$ using the formulas

$$\beta_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}, \quad \beta_0 = \bar{y} - \beta_1\bar{x}.$$

2. Verify that the regression line passes through the point $(\bar{x}, \bar{y})$.
3. Compute the residuals $e_i = y_i - \hat{y}_i$ for each data point.
4. Show that the sum of residuals is zero: $\sum e_i = 0$.

Plot of Data and Regression Line



Problem 11 - Model Performance with R^2

Context: A researcher wants to study the relationship between **Hours Studied (X)** and **Exam Score (Y)** using simple linear regression.

Data (m = 4):

X (Hours)	Y (Score)
1	2
2	4
3	5
4	4

Tasks:

1. Fit the regression line $\hat{Y} = \beta_0 + \beta_1 X$ using the least squares formulas.
2. Compute the **Total Sum of Squares (SST)**:
$$SST = \sum (y_i - \bar{y})^2.$$
3. Compute the **Regression Sum of Squares (SSR)**:
$$SSR = \sum (\hat{y}_i - \bar{y})^2.$$
4. Compute the **Error Sum of Squares (SSE)**:
$$SSE = \sum (y_i - \hat{y}_i)^2.$$

5. Calculate the coefficient of determination:

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}.$$

Interpret its meaning in this context.

Problem 12 - Multivariable Regression Model Representation

Context: Suppose a student's exam score depends not only on **Hours Studied ($\mathbf{X_1}$)**, but also on the **Number of Practice Problems Solved ($\mathbf{X_2}$)**. We want to represent this using a linear regression model.

Data (m = 3):

X_1 (Hours)	X_2 (Problems)	Y (Score)
2	5	70
4	10	88
3	6	78

Tasks:

1. Write the regression model in equation form:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon.$$

2. Express the model for all $m = 3$ data points in **matrix form**:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

3. Identify the dimensions of $\mathbf{Y}$, $\mathbf{X}$, $\boldsymbol{\beta}$, and $\boldsymbol{\varepsilon}$.

Problem 13 - Multivariable Cost Function and Gradient

Problem level - tough

Context: We model a student's exam score Y using two predictors: Hours studied X_1 and Number of practice problems solved X_2. We use a linear model with intercept:

$$\hat{Y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2.$$

Data (m = 3):

X_1	X_2	Y
2	5	70
4	10	88
3	6	78

Tasks:

1. Write the model in matrix form $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta}$ (include the intercept column).
2. Write the cost function (mean squared error) in matrix form:

$$J(\boldsymbol{\beta}) = \frac{1}{2m}\sum_{i=1}^{m}(\hat{y}_i - y_i)^2 \quad \text{or} \quad J(\boldsymbol{\beta}) = \frac{1}{2m}\|\mathbf{X}\boldsymbol{\beta} - \mathbf{Y}\|_2^2.$$

3. Derive the gradient $\nabla_{\boldsymbol{\beta}} J(\boldsymbol{\beta})$ (show matrix result).
4. Evaluate $J(\boldsymbol{\beta})$ at $\boldsymbol{\beta} = \mathbf{0}$ (all zeros).
5. Compute the gradient at $\boldsymbol{\beta} = \mathbf{0}$.
6. Perform one batch gradient descent update from $\boldsymbol{\beta} = \mathbf{0}$ with learning rate $\alpha = 0.001$:

$$\boldsymbol{\beta} \leftarrow \boldsymbol{\beta} - \alpha \nabla_{\boldsymbol{\beta}} J(\boldsymbol{\beta}).$$

   Report the updated $\boldsymbol{\beta}$.

Problem 14 — Multiple Linear Regression (Normal Equation)

Context: Predict **Sales (Y)** from **TV spend ($\mathbf{X_1}$)** and **Radio spend ($\mathbf{X_2}$)**.

Data (n = 4):

X_1 (TV)	X_2 (Radio)	Y (Sales)
1	1	6
2	0	7
0	2	5
3	1	10

Tasks:

1. Write the model: $\hat{Y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2$.
2. Express in matrix form:

$$X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \\ 1 & 3 & 1 \end{bmatrix}, \quad Y = \begin{bmatrix} 6 \\ 7 \\ 5 \\ 10 \end{bmatrix}$$

3. Estimate coefficients using the Normal Equation: $\hat{\beta} = (X^\top X)^{-1} X^\top Y$.
4. Predict Sales for $X_1 = 4,\ X_2 = 2$.

Problem 15 - Underfitting vs. Overfitting

Context: We want to predict exam scores Y from study hours X. We try two models:

- Model A: Linear regression with $\hat{Y} = \beta_0 + \beta_1 X$.
- Model B: 10th-degree polynomial regression with $\hat{Y} = \beta_0 + \beta_1 X + \beta_2 X^2 + \cdots + \beta_{10} X^{10}$.

Observed Training and Test Errors

Model	Training Error (MSE)	Test Error (MSE)
A	12.0	15.0
B	0.5	50.0

Tasks:

1. Which model is likely **underfitting**? Which one is **overfitting**? Justify using the error values.
2. Explain why Model B has very low training error but very high test error.
3. If you wanted a better model, what degree of polynomial might you try? Why?

Problem 16 - Bias–Variance Trade-off

Context: Suppose we train different models on the same dataset. The expected squared error at a test point can be decomposed as:

$$\mathbb{E}\left[(Y - \hat{f}(X))^2\right] = \text{Bias}^2 + \text{Variance} + \sigma^2,$$

where σ^2 is irreducible noise.

Bias–Variance Estimates

Model	Bias^2	Variance	Noise (σ^2)
$SimpleLinear$	25	4	9
$ComplexPolynomial$	4	36	9

Tasks:

1. Compute the total expected error for both models.
2. Which model has higher bias? Which has higher variance?
3. Which model would you prefer in practice, and why?

Problem 17 - Regularization: Ridge vs Lasso (Conceptual)

Context: You have a regression model with many predictors. To avoid overfitting and to handle multicollinearity, you consider regularization.

Tasks:

1. Write down the optimization problems for (a) Ridge regression and (b) Lasso regression.
2. Explain in simple words how Ridge and Lasso differ in their effect on the learned coefficients.
3. Give two situations (short sentences) where you would prefer Ridge over Lasso, and two situations where Lasso is preferable.
4. Briefly explain how the regularization parameter λ affects bias and variance.

Problem 18 - Numerical Example: Ridge Regression

Problem level - tough
Context: A small dataset (no intercept term for simplicity) with two predictors is given. We compare the ordinary least squares (OLS) solution and the Ridge solution.

Data (m = 3)

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad Y = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

(Each row is an observation; columns are two predictors. We set the intercept to zero in this toy example.)

Tasks:

1. Compute $X^\top X$ and $X^\top Y$.
2. Compute the OLS solution $\hat{\beta}^{\text{OLS}} = (X^\top X)^{-1} X^\top Y$.
3. Write the Ridge closed form:

$$\hat{\beta}^{\text{ridge}} = (X^\top X + \lambda I)^{-1} X^\top Y,$$

   and compute $\hat{\beta}^{\text{ridge}}$ for $\lambda = 1$.
4. Compare the OLS and Ridge coefficients; explain the shrinkage effect you observe.